ELECTRONIC SUPPLEMENTARY INFORMATION (ESI)

Predicting Fast-Neutron Permeability Decay and Thermal Dryout Thresholds in Liquid Lithium Capillary Porous Systems

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S1. Mathematical Derivations & First-Principles Proofs

1. **Young-Laplace Capillary Retention:**

10\Delta P_{ ext{cap}} = rac{2 \gamma_{ ext{LV}} \cos heta}{r_p}10

For pristine tungsten mesh saturated with liquid lithium ($\gamma_{ ext{LV}} = 0.350 ext{ N/m}$, $ heta_0 = 15.0^\circ$, {p0} = 25.0\ \mu ext{m}$), $\Delta P_{ ext{cap},0} = 27.05 ext{ kPa}$.

2. **Carman-Kozeny Permeability Model:**

10K = rac{\phi^3}{C_{ ext{CK}} (1-\phi)^2 S_0^2}10

Fast-neutron compaction contracts pore radius (ext{dpa}) = r_{p0} [1 - ■lpha_r (1 - \exp(-■eta_r \cdot ext{dpa}))]$, driving a 0\%$ collapse in permeability at 0 ext{ dpa}$.

3. **Hydrodynamic Friction vs. Capillary Drive:**

The resupply velocity = -(K/\mu_L)

abla P$ drops rapidly despite rising $\Delta P_{ ext{cap}}$, proving that viscous choking in narrowed channels is the governing driver of capillary dryout.

S2. Thermophysical & Material Property Matrix

Parameter / Physical Metric	Design Value	Safety Status / Operational Margin
Pristine Capillary Retention $\Delta P_{ ext{cap}}*	7.05 ext{ kPa}$	Optimal Young-Laplace Head Capacity
Pristine Permeability *	.25 imes 10^{-11} ext{ m}^2$	Carman-Kozeny Baseline Hydraulic Conductivity
Pristine Dryout Heat Flux "_{ ext{dryout}}*	1.50 ext{ MW/m}^2$	(0) = 2.77$ @ 5.0 ext{ MW/m}^2$ Baseline Load
20 dpa Permeability (20)*	-bash.38 imes 10^{-11} ext{ m}^2$	0\%$ Matrix Hydraulic Permeability Collapse
Replacement Limit Threshold	.8 ext{ dpa}*	**Benchmark Safety Limit (= 1.50$, "_{ ext{dryout}} = 22.50 ext{ MW/m}^2$)**

Parameter / Physical Metric	Design Value	Safety Status / Operational Margin
Dryout Parity Limit	3.0 ext{ dpa}*	**Thermal Dryout Limit (= 1.00$, ''_{ ext{dryout}} = 15.00 ext{ MW/m}^2$)**